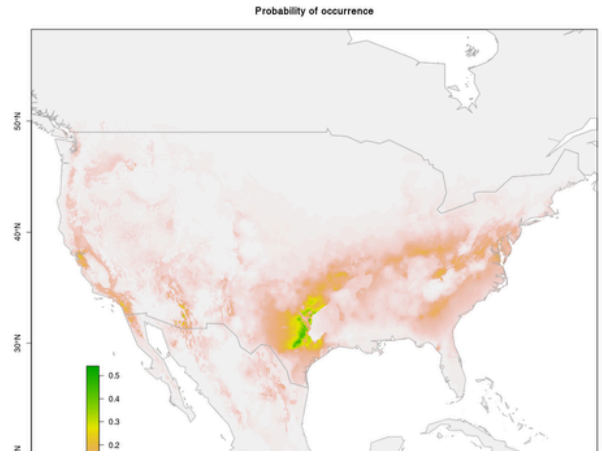
Species Distribution Models



1. Characterize the environmental conditions that are suitable for the species
2. Identify where suitable environments are distributed in space

SO

This does not provide us an ACTUAL distribution

Fundamental Niche

Why might species only occupy part of the Fundamental Niche?

Utility of SDM

- Identifying potential areas for disease outbreak
- Examine niche evolution
- Inform taxonomy
- Conservation
  - reintroduction of species
  - identify priority sites for conservation
  - projecting spread of invasive species and disease

SDM Approaches

Mechanistic Models

- Use physiologically limiting mechanisms to determine ability of species to live in an environment

Required:
detailed understanding of the physiological response of species to environmental factors

Correlative Models

- Estimates environmental conditions suitable for a species by associating species occurrence records with environmental data that likely will affect species physiology and probability of persistence

Required:
Species occurrence records
Environmental variables
Algorithms to connect the two

Data input for Correlative Models

Biological or Geographical Data

- Actual locality data or known distribution of species

Where to get it!

iNaturalist
GBIF
eButterfly
Museum/Herbarium Records
GPS data from field work

Environmental Data

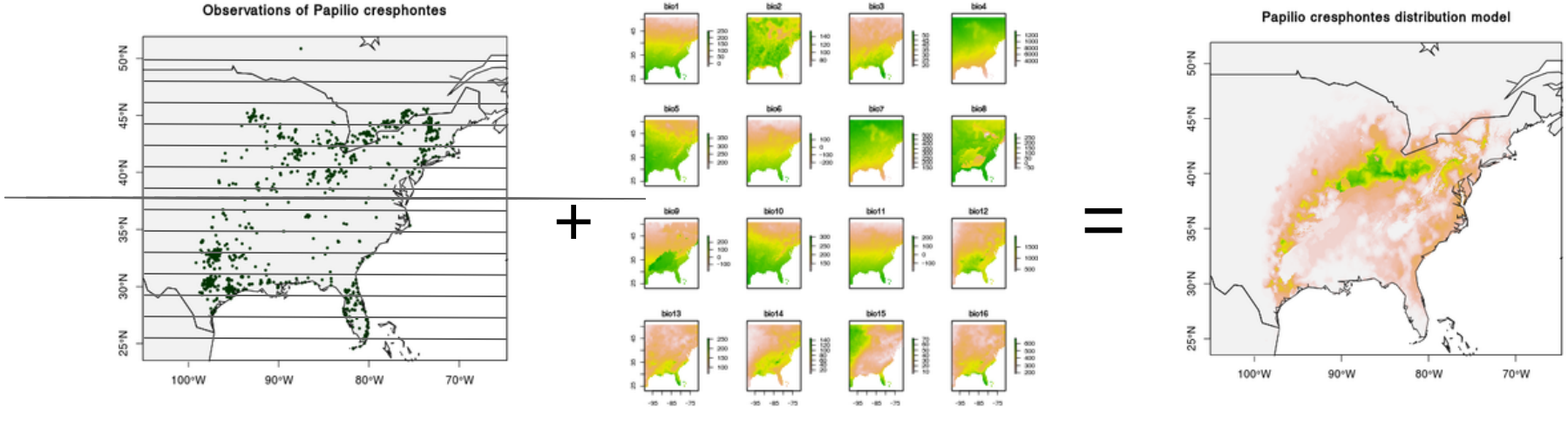
- Used to define the space based on environmental variables (climate, topography, soil type, land cover)
- Set of environmental conditions within which a species can survive and persist

Where to get it!

BioClim
WorldClim

General Steps in SDM

1. Study area is modeled as a raster map composed of grid cells at a specified resolution (for us, North America excluding Mexico)
2. Dependent variables = occurrence records (from iNat)
3. Suite of environmental variables (obtained from BioClim or WorldClim) and used to characterize each cell
4. Function of the environmental variables is generated to predict the degree to which each cell is suitable for the species



Tips for Success Today

- No spaces in file names. NoSpacesInFileNames.doc
- CaPitALIzatiON matters
- Follow the directions carefully!
- Keep all members of the group at the same step - help each other when need be!
- Only CAN-DO attitudes:)
- Patience.

Goal: Complete SP-2 Assignment using instructions handed out today